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Concept Review

Color Sensor

Why Use a Color Sensor?

A color sensor is composed of multiple photodiodes that are color filtered to respond differently to incoming lights. After calibration, these signals from these photodiodes can be used to infer the color of received lights, enabling applications such as intelligent light control, display white balancing, color sorting, and robotics line following.

Color Sensing Principles

A color sensor measures light using photodiodes, semiconductor components that convert incoming photons (light particles) into an electrical current. Because they measure physical quantities of light (not "color" directly), the sensor must rely on optical filters, which are thin films that permit only specific wavelengths to pass through, corresponding to the colors red (~610 nm), green (~ 560 nm) and blue (470 nm), to allow the photodiodes to be sensitive to these wavelengths, as shown in Figure 1. By combining different "colored" photodiodes, color of the coming light can be estimated.

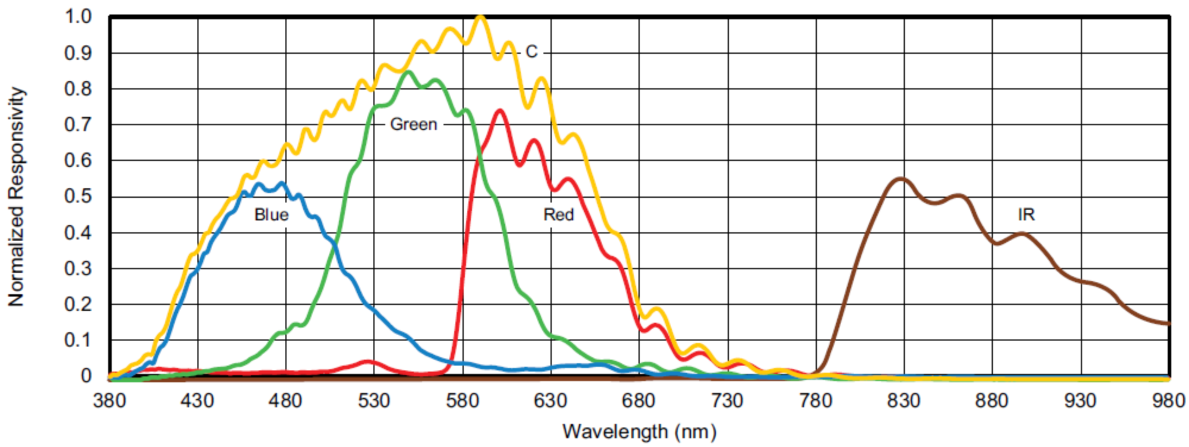

Figure 1. Color sensor (VEML3328) spectral sensitivity

This mechanism closely mirrors the human visual system. Our retinas contain three types of cone cells (L, M, and S) with overlapping spectral sensitivities. Rather than detecting color as fixed categories, the brain interprets color by comparing the relative responses of these three cone types. A RGB color sensor works the same way. Each filtered photodiode sees a different slice of the spectrum, and the combination of R, G, and B responses encodes the perceived color. Many sensors also include a clear (unfiltered) photodiode, which behaves similarly to the eye's rod cells, measuring overall brightness and improving sensitivity under low-light conditions.

To decode the color perceived by the sensor, we need to convert it into a universal, device-independent color space. This is where the **CIE 1931 XYZ tristimulus system** comes in, that defines color by XYZ coordinates based on standardized human visual response curves, allowing any device's raw measurements to be mapped into a common standard of color. In the next section, we will explore how these tristimulus values are defined and what it means for modern digital devices.

Standardization of Color

In the 1920s, scientists W. D. Wright and J. Guild conducted foundational experiments to better understand how humans perceive color. In their studies, observers viewed a test light on one side of a partitioned box and adjusted three monochromatic primary lights (red, green, blue) on the other side until the mixture visually matched the test light, as illustrated in Figure 2.

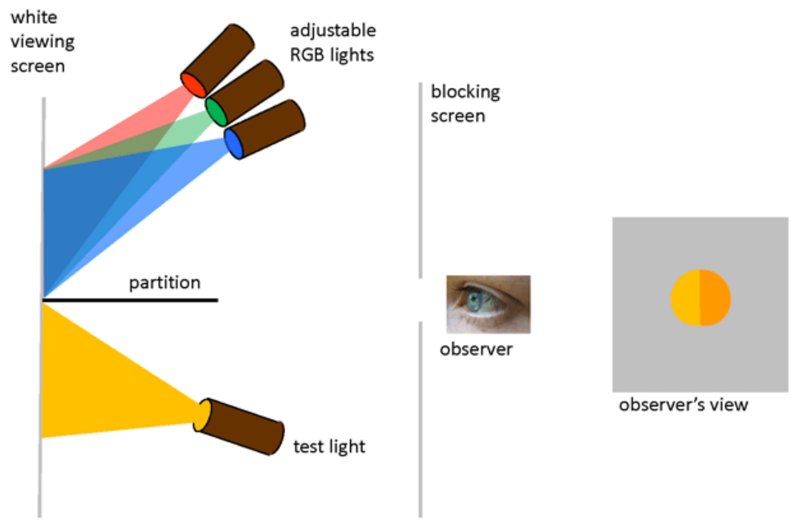

Figure 2. Wright-Guild Experiment on Color Perception

As the wavelength of the test light was swept across the visible spectrum, researchers recorded how much of each primary was needed to create a match. These measurements formed the original RGB color matching functions. Then in 1931, The International Commission on Illumination (CIE) standardized these functions, and transformed the original RGB color functions into three new components, X, Y, and Z, known as **CIE XYZ tristimulus values**, shown in figure 3.

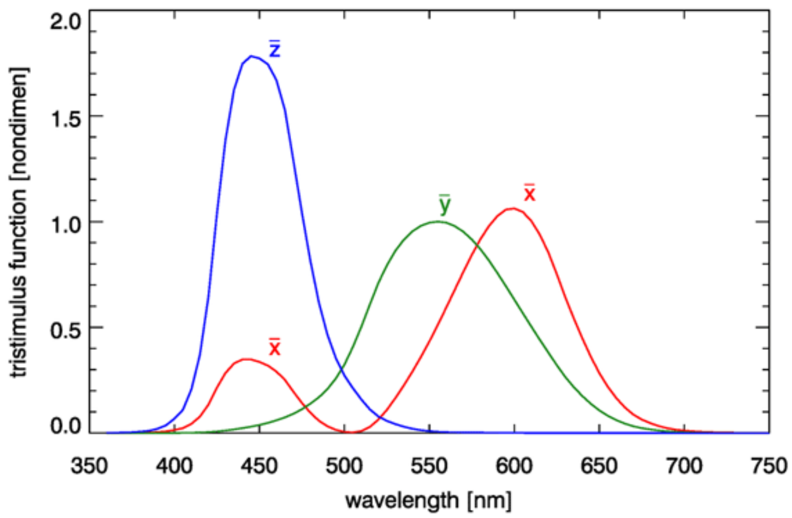

Figure 3. CIE 1931 colour-matching functions (2 degree observer).

Each tristimulus value represents how much of the standardized X, Y, and Z values are needed to describe a given color. Because XYZ values scale with the overall brightness of the light source, they are often normalized to remove dependency of the light intensity. This yields the chromaticity coordinates:

$$(x, y, z) = \left(\frac{X}{X+Y+Z}, \frac{Y}{X+Y+Z}, \frac{Z}{X+Y+Z}\right)$$

Since $x + y + z = 1$, only two coordinates are needed to describe a chromaticity. By conventions, the x and y values are used, which allows colors to be plotted on a two-dimensional plane. Mapping these coordinates creates the CIE chromaticity diagram, as shown in Figure 4.

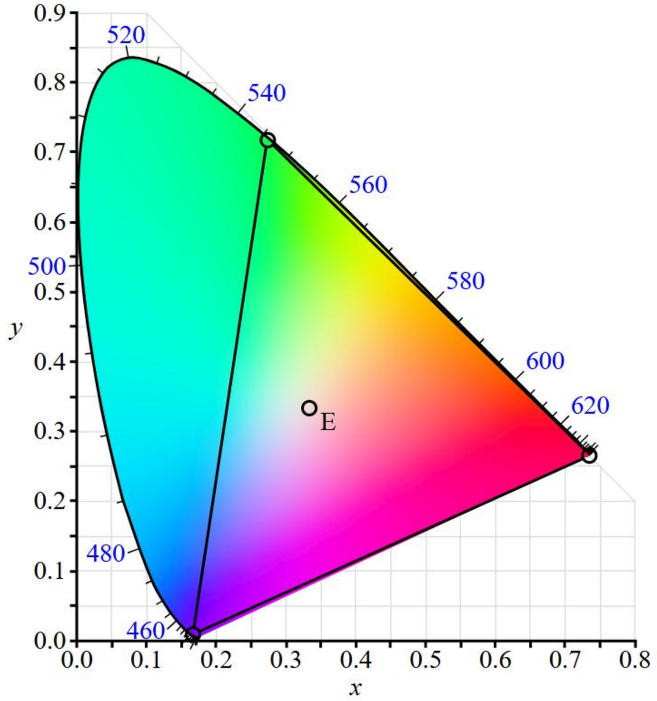

Figure 4. The CIE 1931 2 degree chromaticity diagram.

The outer boundary of this plot, representing pure monochromatic wavelengths, is called the **spectral locus**. Every point inside the boundary represents a visible color created by mixing two or more wavelengths. This color space provides a standardized way to describe colors numerically, ensuring consistent communication across displays, cameras, and lighting systems.

Selecting any two points within the color space defines a line, and every point along that line represents a mixture of those two colors. A **color gamut**, most commonly a triangle inside the chromaticity diagram, represents the set of all possible colors that can be produced as a mixture of the vertices, which is known as the **primary colors** of the gamut. For example, if the primaries of an RGB display are red (≈650 nm), green (≈550 nm), and blue (≈450 nm), a triangle on the chromaticity diagram can be formed as shown in figure 4. All colors the display can produce lie within that triangle. In reality, display primaries are not perfectly monochromatic, so the actual primaries lie inside the spectral locus. As a result, real display gamuts cover smaller than the idealized spectral triangle shown in Figure 4.

Different display technologies use different RGB primaries, with the most common being **sRGB**, which is used in standard consumer monitors. UHD systems use the Rec. 2020 color space, whose primaries lie closer to the spectral locus and therefore produce a much wider and more vibrant color gamut. Emerging display technologies can use more than three primaries to achieve extremely wide gamut beyond conventional RGB limitations.

Datasheet and Considerations

Determining the requirements for a suitable color sensor involves considering several factors and consulting the datasheet for key specifications. Like other optical sensors, consider the field of view of a color sensor, which indicates the size and shape of the measurement area. Field of view requirements are determined by where the target objects will be located relative to where the sensor will be mounted. The amount of ambient light in the operating environment can also impact how well the sensor can distinguish between different colors. Additional lighting sources or shielding can be added to the system to ensure consistent lighting conditions.